IdiomX: A Bilingual Benchmark for Idiom Understanding with Hybrid Generation–Retrieval Modeling

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Abstract

Idiomatic expressions pose a significant challenge in natural language processing due to their non-literal and context-dependent meanings. In this paper, we introduce IdiomX, a large-scale, high-quality bilingual dataset designed for idiom detection, interpretation, and cross-lingual understanding. The dataset contains over 123,000 contextualized examples derived from approximately 15,000 English idioms, each annotated with idiomatic and literal usage, semantic features, and English–Arabic translations.

To ensure quality and scalability, we employ a hybrid pipeline combining curated lexical resources with large language model (LLM)-based enrichment. The dataset undergoes rigorous validation, achieving over 99.98% label consistency and near-complete bilingual coverage.

Beyond dataset construction, we benchmark IdiomX on idiom detection and context-to-idiom generation. Transformer-based models substantially outperform lexical baselines on idiom detection, while a proposed generation-first hybrid model achieves the best performance on context-to-idiom generation. These findings show that idiom understanding is not purely a similarity or classification problem, but benefits from combining contextual generation and retrieval-based grounding.

Keywords: idioms, NLP, bilingual dataset, semantic understanding, LLM, Arabic NLP

I. Introduction

Idiomatic expressions are a fundamental component of natural language, often conveying meanings that cannot be inferred directly from the individual words. This non-compositional nature makes idioms particularly challenging for natural language processing (NLP) systems, especially in tasks such as machine translation, semantic understanding, and information retrieval.

Despite their importance, existing idiom datasets are typically limited in size, lack contextual diversity, or are restricted to a single language. Furthermore, most available resources do not provide rich annotations such as semantic consistency, compositionality, or multilingual mappings.

To address these limitations, we present IdiomX, a large-scale bilingual dataset for idiomatic expression understanding. The dataset combines curated lexical sources with automated large language model (LLM) enrichment to generate high-quality contextual examples and annotations. In addition to English idioms, IdiomX provides aligned Arabic translations, enabling cross-lingual research.

The main contributions of this paper are:

1. We introduce IdiomX, a large-scale bilingual dataset for idiom understanding with 123,336 contextualized examples and 14,986 idioms.
2. We provide balanced idiomatic and literal contextual usage suitable for controlled evaluation.
3. We release English–Arabic aligned idiom data for cross-lingual idiom research.
4. We benchmark the dataset on idiom detection and context-to-idiom generation.
5. We show that a generation-first hybrid approach outperforms both retrieval-only and generation-only baselines on context-to-idiom prediction.

This work contributes not only a dataset, but also empirical insights into how idiomatic meaning can be modeled across languages.

II. Related Work

Several datasets have been proposed for idiom detection and interpretation, including resources derived from lexical databases such as WordNet and Wiktionary. While these datasets provide useful foundations, they often lack contextual diversity and large-scale annotations.

Existing idiom datasets are typically limited in scale, linguistic diversity, or annotation depth. To better position IdiomX within the current landscape, we provide a quantitative comparison with widely used idiom datasets in Table 1.

Dataset	Language	Idioms	Instances	Bilingual	Link
VNC-Tokens	EN	53	2,984	No	aclanthology
SemEval-2013 Task 5b	EN	85	~4,350	No	aclanthology
IDIX	EN	52	4,022	No	aclanthology
MAGPIE	EN	~1,700	56,000	No	aclanthology
PIE-English	EN	~1,200	~20,100	No	arxiv
IdiomX (Ours)	**EN+AR**	**14,986**	**123,336**	Yes	GitHub

As shown in Table 1, existing idiom datasets are significantly smaller in both scale and linguistic diversity. Most datasets focus on a limited number of idiom types and do not provide bilingual coverage. In contrast, IdiomX contains over 123,000 contextualized examples across nearly 15,000 idioms with aligned English–Arabic representations and rich semantic annotations.

Recent work has explored the use of neural models for idiom understanding, leveraging transformer-based architectures such as BERT and T5. However, the effectiveness of these models is often constrained by the availability of high-quality training data.

In multilingual settings, idiom datasets remain particularly scarce, with limited resources supporting cross-lingual analysis. This gap highlights the need for comprehensive datasets that combine scale, contextual richness, and multilingual support.

IdiomX addresses these challenges by providing a large-scale, bilingual dataset enriched with semantic annotations and validated using automated and rule-based techniques

III. DATASET CONSTRUCTION

The IdiomX dataset is constructed through a multi-stage pipeline combining data collection, preprocessing, enrichment, and validation.

First, idiomatic expressions are collected from multiple lexical sources, including Wiktionary and WordNet. Duplicate entries and low-quality records are removed through normalization and filtering.

Next, contextual examples are generated and enriched using a large language model (LLM). For each idiom, multiple example sentences are produced, covering both idiomatic and literal usage. The enrichment process also generates English and Arabic meanings, as well as aligned translations of the example sentences.

To ensure consistency, additional linguistic features are computed, including compositionality level, learner difficulty, and semantic similarity scores. These features provide deeper insights into idiomatic behavior and support advanced modeling tasks.

The final dataset consists of over 123,000 examples with rich annotations and bilingual coverage.

The overall dataset construction pipeline is illustrated in Figure 1.

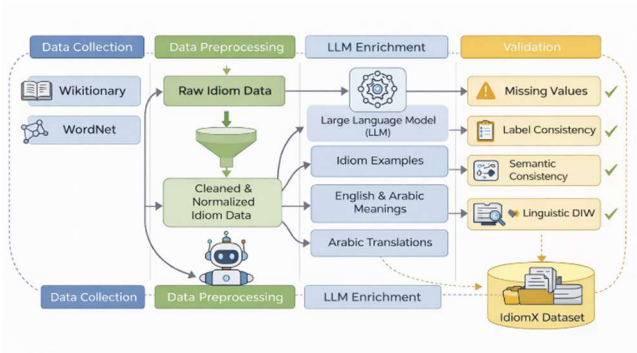

Figure 1: Overview of the IdiomX dataset construction pipeline

A. *Annotation and Validation Strategy*

Unlike traditional datasets relying on manual annotation by linguistic experts, IdiomX leverages LLM-assisted annotation combined with automated validation and targeted manual inspection. This approach enables large-scale dataset construction while maintaining high consistency and semantic alignment.

The LLM is prompted to generate structured outputs including contextual examples, idiomatic usage labels, semantic meanings, and bilingual translations. To ensure quality, the pipeline enforces strict schema constraints and performs multiple validation steps.

A subset of samples is manually reviewed to verify correctness, ensuring alignment between idiomatic usage and contextual meaning. This hybrid approach enables scalability while maintaining high data quality.

B. *LLM-based enrichment and Prompting Strategy*

LLM-based enrichment is performed using structured prompts to generate multiple examples per idiom, covering both idiomatic and literal usage. Each example includes semantic meanings in English and Arabic, along with additional linguistic features such as compositionality and difficulty level.

The generation process is controlled to ensure consistency and diversity across samples, followed by a verification stage to detect and correct potential errors.

The dataset enrichment process is guided by structured prompting techniques to ensure consistent and high-quality outputs. The LLM is instructed to generate multiple examples per idiom, covering both idiomatic and literal usage, along with semantic meanings and bilingual translations.

The prompt enforces a fixed output schema, ensuring that each generated instance includes required fields such as idiom surface form, contextual usage, semantic meaning, and usage label.

To reduce variability and improve reliability, prompts are designed with explicit instructions, constraints, and examples. A verification stage is applied to refine outputs and correct inconsistencies.

Full prompt templates and generation scripts are provided in the accompanying repository for reproducibility.

Unless otherwise noted, automatically generated fields are subsequently validated through schema checks, semantic consistency checks, and targeted manual inspection.

IV. Data Validation & Quality

To ensure the reliability of the dataset, multiple validation steps are applied.

Missing values are analyzed across all fields, with critical attributes such as contextual examples and meanings exhibiting negligible missing rates (<0.01%). Label consistency between categorical annotations and boolean indicators exceeds 99.98%, indicating highly reliable labeling.

Surface-form alignment between idioms and their occurrences in context is evaluated using both strict and relaxed matching strategies, achieving high alignment rates.

Additionally, semantic consistency between idioms and their contextual usage is measured using embedding-based similarity metrics. The dataset achieves a mean semantic consistency score of 0.59, reflecting strong alignment while preserving natural variation.

Validation combines rule-based checks, consistency checks across paired fields, lexical alignment checks, and sample-based manual review.

V. Dataset Analysis

The IdiomX dataset contains approximately 14,986 unique idioms, with an average of 8.2 contextual examples per idiom. This provides both diversity and depth for modeling.

Example sentence lengths are well-balanced, with an average of approximately 12.5 words per example, ensuring sufficient contextual information without excessive noise.

The dataset is evenly balanced between idiomatic and literal usage, with each category representing approximately 50% of the data. This balance is critical for training robust classification models.

Furthermore, IdiomX provides near-complete bilingual coverage, with over 99.99% of examples and meanings available in both English and Arabic. This enables cross-lingual research and applications in multilingual NLP.

Overall, the dataset demonstrates strong statistical properties, linguistic diversity, and high-quality annotations.

Idioms in the dataset vary in compositionality, ranging from fully non-compositional expressions (e.g., "kick the bucket") to partially compositional idioms. This variation introduces different levels of semantic difficulty, making IdiomX suitable for evaluating models across a spectrum of linguistic complexity.

Metric	Value
Total examples	123,336
Unique idioms	14,986
Avg examples/idiom	8.2
Arabic coverage	99.99%
Label balance	50/50

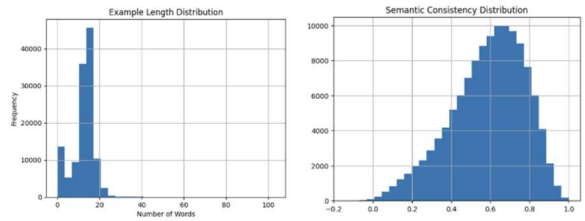

Figure 2: Distribution of sentence lengths and semantic consistency scores in IdiomX.

IdiomX captures a wide range of idiomatic expressions, including metaphorical, compositional, and partially compositional idioms. The dataset also reflects varying levels of semantic transparency and ambiguity, which are critical for evaluating idiom understanding systems.

Additionally, the dataset includes idioms from different domains and registers, providing diverse linguistic coverage for both formal and informal language.

VI. Benchmark Tasks

IdiomX supports multiple downstream NLP tasks involving idiomatic language.

A. *Task 1: Idiom Detection*

B. *Task 2: Context-to-Idiom Generation*

C. *Task 3: Arabic Context-to-Idiom Generation*

These tasks are enabled by structured dataset fields such as idiom_in_example, idiom_in_example_meaning_en, idiom_in_example_meaning_arabic, and is_example_idiom.

VII. Experiments

To evaluate the effectiveness of IdiomX, we conduct baseline experiments across multiple tasks supported by the dataset.

A. *Task 1: Idiom Detection*

We formulate idiom detection as a binary classification task, where the goal is to determine whether a given sentence contains an idiomatic expression or a literal usage.

Formally, given an input sentence $x = (w_1, w_2, ..., w_n)$, the goal is to predict a binary label $y \in \{0,1\}$, where $y = 1$ indicates idiomatic usage and $y = 0$ indicates literal usage.

1. *Formulation*

The model learns a function:

$\hat{y} = f(x; \theta)$, where θ represents model parameters.

The Logistic Regression model estimates the probability of idiomatic usage:

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

where σ is the sigmoid function.

The model is trained using binary cross-entropy loss:

$$L = -\Sigma [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

where $\hat{y}$ is the predicted probability.

Transformer-based models compute contextual representations using self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

where Q, K, and V are query, key, and value matrices.

The final prediction is obtained by applying a classification layer on top of the [CLS] token representation:

$$\hat{y} = \text{softmax}(W\ h_cls + b)$$

2. *Models*

We evaluate a range of models spanning classical and transformer-based approaches:

- Logistic Regression with TF–IDF features (baseline)
- Tuned Logistic Regression
- RoBERTa (base)
- DeBERTa-v3 (base)

3. *Results*

Compare surface-level lexical models against deep contextual representations.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8406	0.84	0.84	0.844
Logistic Regression (Tuned)	0.8435	0.85	0.84	0.846
RoBERTa	~0.90	0.9	0.9	0.902
DeBERTa	**0.91**	**0.91**	**0.91**	**0.909**

Main Results table

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

Performance differences between models are consistent across validation splits

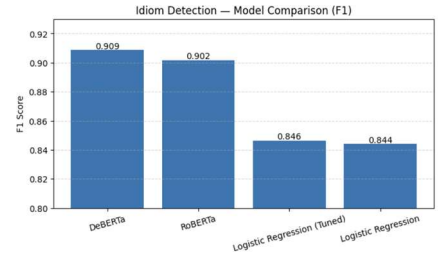

Figure 3: Model comparison on the idiom detection task (F1 score)

The large performance gap (~ 6-7%) between classical and transformer models demonstrates that idiom detection is fundamentally a semantic task requiring contextual understanding rather than surface lexical matching.

4. *Calibration and confidence*

The confusion matrix for DeBERTa shows balanced performance across both classes, with 4861 correct literal predictions and 5014 correct idiomatic predictions.

The model produces 517 false positives and 491 false negatives, indicating no significant class bias.

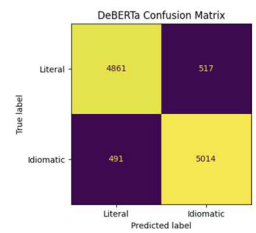

Figure 4: DeBERTa Confusion Matrix

ROC + PR CURVES

DeBERTa achieves an AUC of 0.968, significantly outperforming the baseline (0.915), indicating strong discriminative ability across decision thresholds.

Similarly, the precision–recall curve shows that DeBERTa maintains high precision across recall levels, confirming robustness in identifying idiomatic expressions.

The Area Under the ROC Curve (AUC) measures the probability that the model ranks a randomly chosen positive instance higher than a negative one.

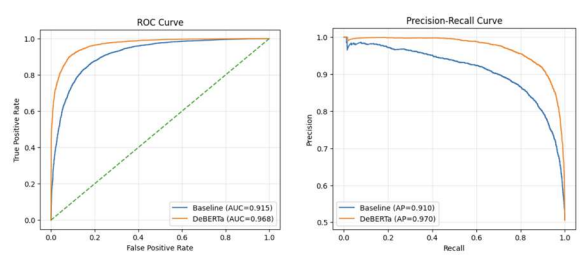

Figure 5: ROC + PR CURVES

CONFIDENCE DISTRIBUTION

The confidence distribution exhibits a clear bimodal pattern, with predictions concentrated near 0 and 1.

This indicates strong class separation and well-calibrated model behavior, with minimal uncertainty in most predictions.

DeBERTa shows slightly sharper confidence peaks compared to RoBERTa, suggesting more decisive predictions.

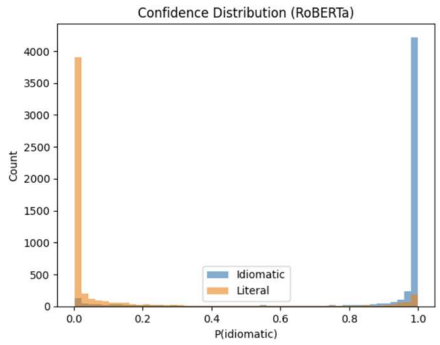

Figure 6: RoBERTa confidence

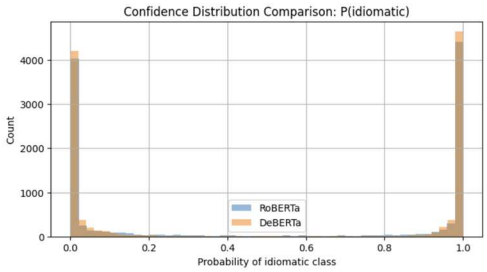

Figure 7: DeBERTa comparison

Well-calibrated models satisfy: $P(y = 1 \mid p \approx 0.8) \approx 0.8$

Both models show strong class separation, with probabilities concentrated near 0 and 1. DeBERTa exhibits slightly sharper confidence, indicating more decisive predictions.

DeBERTa produces the lowest number of misclassifications (1008), followed by RoBERTa (1085), while the baseline model shows significantly higher error rates (1735).

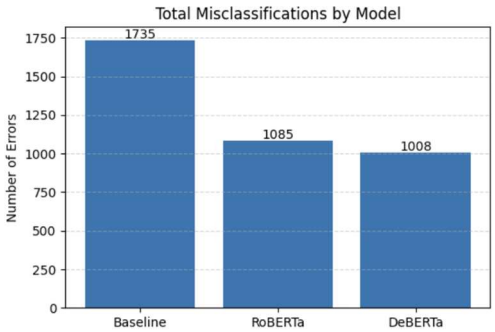

Figure 8: Error count comparison

5. ***Error analysis reveals several key failure modes***:

(1) Short or context-limited inputs:

Examples such as “baby sister” or “Cuban bread” lack sufficient context for reliable classification.

(2) Named entities and domain-specific expressions:

Phrases like “Benjamin Button” or “Bravo Zulu” are often misclassified due to ambiguity between literal and idiomatic usage.

(3) Context-dependent idioms:

Expressions such as “one for the road” require broader discourse context to be correctly interpreted.

(4) Ambiguous phrases:

Certain expressions can be interpreted both literally and idiomatically depending on context, representing inherent challenges for the task.

Notably, some errors occur with high confidence, indicating that certain ambiguous cases remain challenging even for advanced models.

Analysis of prediction confidence shows that correct predictions are typically associated with very high confidence, while incorrect predictions tend to occur at lower confidence levels.

DeBERTa outperforms RoBERTa due to its disentangled attention mechanism, which separately models content and positional information.

6. *Summary*

Overall, the results demonstrate that idiom detection is a challenging semantic task that cannot be effectively solved using surface-level lexical features alone.

Transformer-based models, particularly DeBERTa, achieve strong performance by leveraging contextual representations, enabling accurate distinction between idiomatic and literal expressions.

However, remaining errors highlight the inherent ambiguity of idiomatic language and the need for deeper contextual and world knowledge in future work.

This allows the model to better capture subtle contextual cues and semantic relationships, which are critical for idiom understanding.

This suggests that model uncertainty correlates with semantic ambiguity in the data.

B. ***Task 2: Context-to-Idiom Generation***

In this task, the model is given a contextual sentence or semantic description and is required to predict the most appropriate idiomatic expression. Unlike idiom detection, this task requires both semantic understanding and generation capability, making it more challenging and closer to real-world language use.

Formally: $y^* = \text{argmax}_y P(y \mid x)$
where y is selected from a predefined idiom inventory.

A. *SBERT Retrieval Model*

Each context x and candidate idiom i are encoded into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$

Similarity is computed using cosine similarity:
$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \, \|h_i\|)$

Prediction: $\hat{y} = \text{argmax}_i \, \text{sim}(x, i)$

B. FLAN-T5 Generative Model

The model generates an idiom sequence y autoregressively:

$P(y \mid x) = \prod P(y_t \mid y_{<t}, x)$

Prediction: $\hat{y} = \text{argmax}_y \, P(y \mid x)$

C. Proposed FLAN-First Hybrid Model

"$\hat{y}$ equals y_gen if Match $\geq \tau$, otherwise the best retrieval candidate."

$\text{Match}(y_gen, I) = \max_i \, \text{similarity}(y_gen, i)$

1. *Experimental setup*

$$\text{Top-K Accuracy} = (1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$$

$$\text{MRR} = (1/N) \sum (1 / \text{rank}_i)$$

2. *Quantitative Results:*

Model	Exact Match (Top-1)	Top-3	Top-5	MRR@100
SBERT Retrieval	0.461	0.599	0.65	0.549
FLAN-T5	0.678			
Hybrid FLAN-first	0.818			

Performance differences between models are consistent across validation splits

The results show that:

SBERT provides strong semantic retrieval but suffers from ranking limitations.

FLAN-T5 improves contextual reasoning but may produce invalid or noisy outputs.

The hybrid model combines generative reasoning with retrieval grounding, improving both accuracy and robustness.

This confirms that idiom prediction benefits from a two-stage reasoning process, where generation captures semantic intent and retrieval enforces discrete correctness.

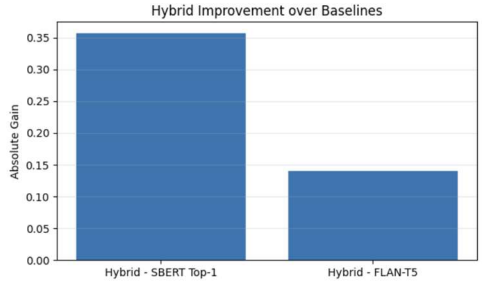

Figure 9: Hybrid Improvement over Baselines

Comparison	Gain
Hybrid vs SBERT	0.356
Hybrid vs FLAN	0.14

The hybrid model improves exact-match accuracy by 35.7 percentage points over SBERT Top-1 and by 14.0 points over FLAN-T5.

Decision	Count	Percent
FLAN exact bank	7885	0.724458
FLAN fuzzy correction	608	0.055862
SBERT fallback	2391	0.21968

3. *Hybrid Decision Analysis*

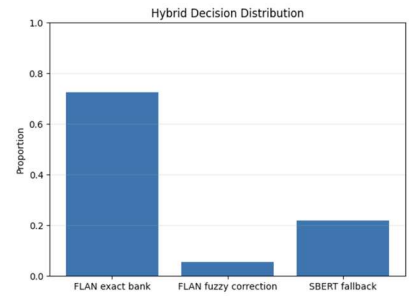

Figure 10: Hybrid Decision Distribution

Most predictions are resolved by the generative model, while retrieval provides corrective support.

4. *Error Analysis*

Errors are categorized into:
1. Retrieval Error: correct idiom not retrieved
2. Ranking Error: correct idiom not ranked first
3. Generation Error: incorrect or noisy output

The hybrid model reduces both ranking and generation errors

Model Interaction Analysis

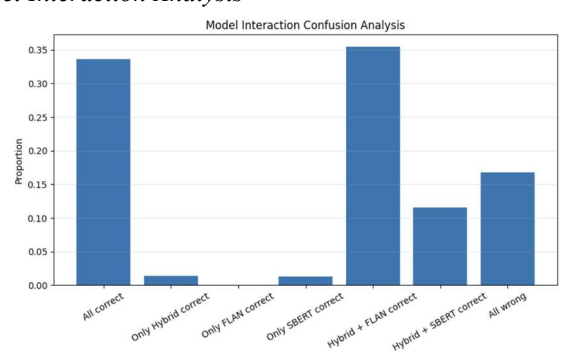

Figure 11: Model Interaction Confusion chart

These results indicate that the hybrid model is not merely averaging predictions, but recovering additional correct cases through complementary decision behavior.

5. *Qualitative Analysis*

Case Study 1: Hybrid corrects both models

- SBERT retrieval fails to select the correct idiom
- FLAN generates an incorrect or approximate phrase
- The hybrid model successfully corrects the output

Case Study 2: FLAN succeeds alone

- FLAN correctly generates the idiom directly
- SBERT retrieval may rank incorrect candidates
- The hybrid model accepts the FLAN output without modification

Case Study 3: SBERT fallback saves FLAN

- FLAN produces incorrect or incomplete outputs
- SBERT retrieval successfully identifies the correct idiom
- The hybrid model switches to retrieval-based prediction

These examples demonstrate complementary strengths between generation and retrieval.

6. *Discussion*

The results indicate that idiom prediction is not purely a retrieval task, but benefits from generative reasoning.

The hybrid model effectively integrates both paradigms, improving robustness and accuracy.

The hybrid approach introduces additional computational cost due to dual processing, but this is justified by the performance gain.

This improvement arises because generative models capture semantic flexibility, while retrieval models enforce discrete correctness within a constrained idiom inventory

7. *Task Summary*

The proposed hybrid model achieves the best performance on the context-to-idiom task.

By combining generative reasoning with retrieval-based grounding, it overcomes the limitations of both approaches and provides a robust solution for idiomatic understanding.

C. *Task 3: Arabic Context-to-Idiom (planned benchmark)*

This task extends context-to-idiom generation to a cross-lingual setting. Given an Arabic
sentence or meaning, the model generates the corresponding English idiom.

This task evaluates the ability of models to capture semantic alignment across languages,
leveraging the bilingual structure of IdiomX.

The task is modeled similarly to Task 2 using a sequence-to-sequence transformer, with evaluation
based on generation quality and semantic correctness.

VIII. Discussion

The experimental results demonstrate that IdiomX supports a wide range of NLP tasks with meaningful performance.

The experimental findings highlight three key insights. First, idiom detection requires contextual modeling rather than surface-level lexical matching. Second, context-to-idiom generation is not purely a retrieval problem, but requires semantic reasoning. Third, hybrid approaches outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

The experiments suggest that idiom understanding benefits from a division of labor: generative models provide flexible contextual reasoning, while retrieval mechanisms provide discrete grounding and correction.

IX. Applications

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- idiom detection
- idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language

X. Dataset Availability

The IdiomX dataset is publicly available via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration

XI. Limitations

While IdiomX provides large-scale coverage and rich annotations, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Arabic translations primarily reflect standard written Arabic and may not fully capture dialectal variations.

Future work includes incorporating human annotation and extending the dataset to additional languages.

Some generated idiomatic or literal examples may reflect model-preferred phrasing rather than naturally occurring corpus frequency.

Additionally, evaluation for the generation task relies primarily on exact match metrics, which may not fully capture semantic equivalence between idioms.

XII. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding. By combining curated sources with LLM-based enrichment and rigorous validation, the dataset achieves high quality, strong semantic alignment, and near-complete bilingual coverage.

IdiomX addresses key limitations in existing resources and provides a comprehensive benchmark for idiom-related tasks. Future work includes extending the dataset to additional languages and exploring advanced modeling approaches using the dataset.

Beyond dataset construction, benchmark experiments demonstrate that transformer-based models outperform lexical baselines on idiom detection, and that a generation-first hybrid strategy provides the strongest performance on context-to-idiom generation.

These results position IdiomX as a valuable benchmark for advancing research in figurative language understanding and cross-lingual semantic modeling.

[1] Devlin et al. (2019), BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. BERT learns deep bidirectional contextual representations via masked language modeling and next-sentence prediction, and can be fine-tuned for downstream classification tasks. (https://aclanthology.org/N19-1423.pdf)

[2] Raffel et al. (2020), T5: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. T5 casts all NLP tasks as text-to-text generation, which makes it a natural model family for context-to-idiom and cross-lingual generation tasks. (https://arxiv.org/pdf/1910.10683)

[3] Haagsma et al. (2020), MAGPIE. A large corpus of potentially idiomatic expressions; especially useful for comparison with English contextual idiom datasets. (https://aclanthology.org/2020.lrec-1.35.pdf)

[4] Sporleder et al. (2010), Idioms in Context: The IDIX Corpus. Contextual idiom disambiguation resource. (https://aclanthology.org/L10-1425/)

[5] Korkontzelos et al. (2013), SemEval-2013 Task 5b. Ambiguous idioms in context benchmark. (https://aclanthology.org/S13-2007.pdf)

[6] Adewumi et al. (2022), PIE-English. A larger English corpus with multiple figurative classes. (https://arxiv.org/abs/2105.03280)

[7] Salton et al. (2016), Idiom Token Classification using Sentential Distributed Semantics. Uses VNC-Tokens and provides one of the classic contextual idiom classification baselines. (https://aclanthology.org/P16-1019.pdf)